Property	$x(t) \overset{FT}{\longleftrightarrow} X(\Omega) \quad y(t) \overset{FT}{\longleftrightarrow} Y(\Omega)$
Linearity	$ax(t) + by(t) \overset{FT}{\longleftrightarrow} aX(\Omega) + bY(\Omega)$
Time Shift	$x(t - t_o) \overset{FT}{\longleftrightarrow} e^{-j\Omega t_o} X(\Omega)$
Frequency Shift	$e^{j\gamma t} x(t) \overset{FT}{\longleftrightarrow} X(\Omega - \gamma)$
Scaling	$x(at) \overset{FT}{\longleftrightarrow} \frac{1}{\lvert a \rvert} X\left(\frac{\Omega}{a}\right)$
Differentiation in Time	$\frac{d}{dt} x(t) \overset{FT}{\longleftrightarrow} j\Omega X(\Omega)$
Differentiation in Frequency	$-jtx(t) \overset{FT}{\longleftrightarrow} \frac{d}{d\Omega} X(\Omega)$
Integration	$\int_{-\infty}^{t} x(\tau) d\tau \overset{FT}{\longleftrightarrow} \frac{X(\Omega)}{j\Omega} + \pi X(0)\delta(\Omega)$
Convolution-Multiplication	$\int_{-\infty}^{\infty} x(\tau) y(t - \tau) d\tau \overset{FT}{\longleftrightarrow} X(\Omega) Y(\Omega)$
Multiplication-Convolution	$x(t)y(t) \overset{FT}{\longleftrightarrow} \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\Gamma) Y(\Omega - \Gamma) d\Gamma$
Parseval's Theorem	$\int_{-\infty}^{\infty} \lvert x(t) \rvert^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} \lvert X(\Omega) \rvert^2 d\Omega$
Symmetry (for $x(t)$ real)	$X(-\Omega) = X^*(\Omega)$ $\text{Real}\{X(\Omega)\} = \text{Real}\{X(-\Omega)\}$ $\text{Imag}\{X(\Omega)\} = -\text{Imag}\{X(-\Omega)\}$ $\lvert X(\Omega) \rvert = \lvert X(-\Omega) \rvert$ $\arg\{X(\Omega)\} = -\arg\{X(-\Omega)\}$